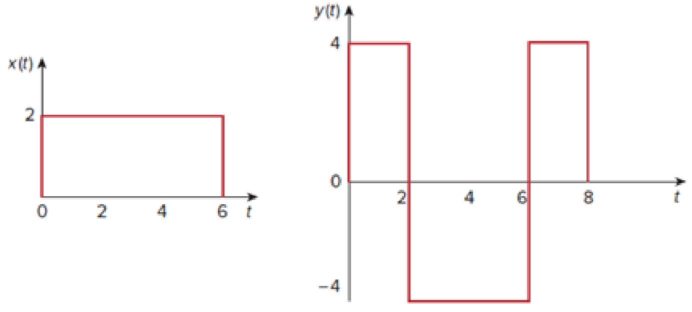
Problem

Let $x(t)$ and $y(t)$ be as shown in Fig. 15.36. Find $z(t) = x(t) * y(t)$.

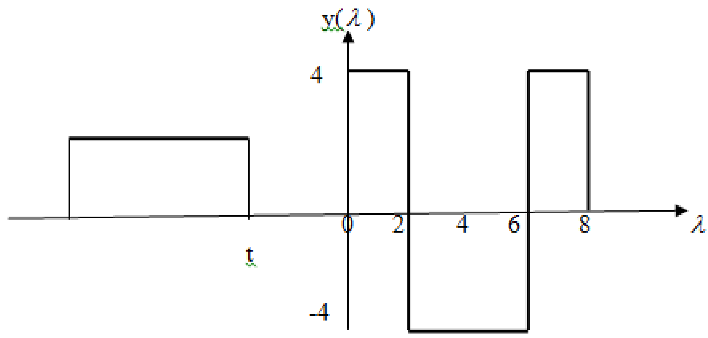
Figure 15.36:



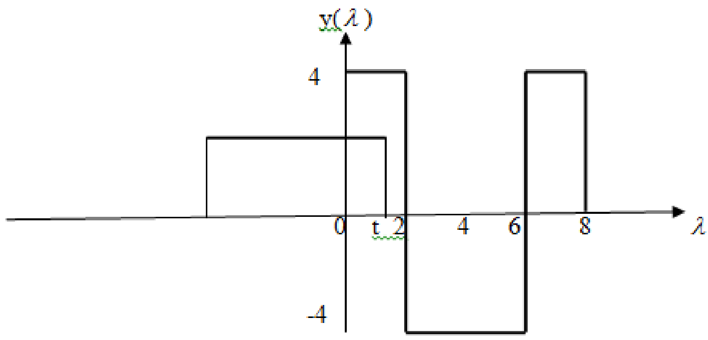
Step-by-step solution

Step 1 of 1

We fold x(t) and slide on y(t). For t>0, no overlapping as shown below. x(t) =0.

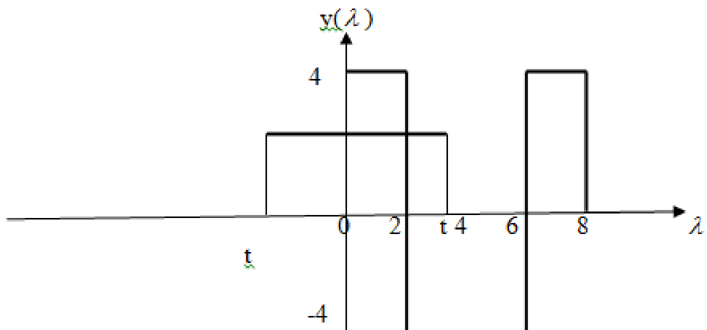


For 0<t<2, there is overlapping, as shown below.



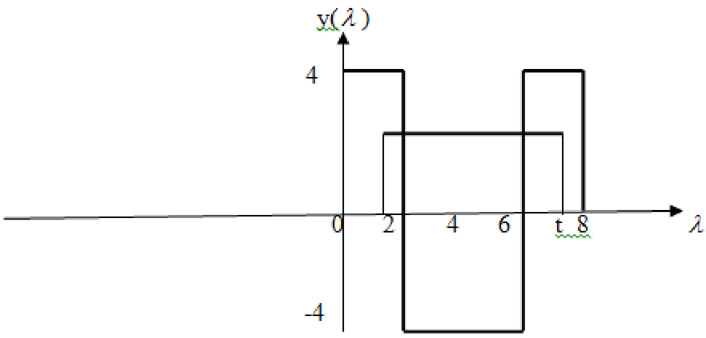
$$z(t) = \int_0^t (2)(4)dt = 8t$$

For 2<t<6, the two functions overlap, as shown below.



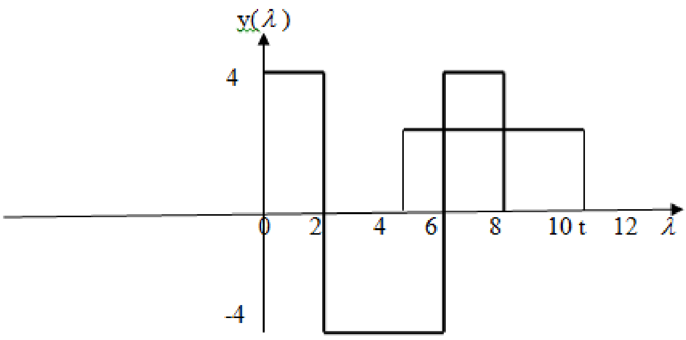
$$z(t) = \int_0^2 (2)(4)d\lambda + \int_0^t (2)(-4)d\lambda = 16 - 8t$$

For 6>t>8, they overlap as shown below.



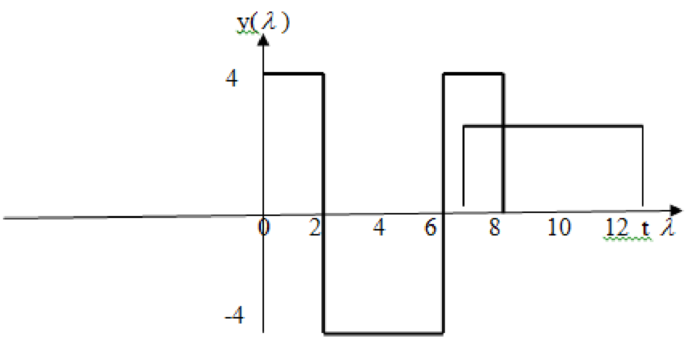
$$z(t) = \int_{t-6}^{2} (2)(4)d\lambda + \int_2^6 (2)(-4)d\lambda + \int_6^t (2)(4)d\lambda = 8\lambda\Big|_{t-6}^{2} - 8\lambda\Big|_2^6 + 8\lambda\Big|_6^t = -16$$

For 8> t >12, they overlap as shown below.



$$z(t) = \int_{t-6}^{6} (2)(-4)d\lambda + \int_6^8 (2)(4)d\lambda = -8\lambda\Big|_{t-6}^{6} + 8\lambda\Big|_6^8 = 8t - 80$$

For 12<t<14, they overlap as shown below.



$$z(t) = \int_{t-6}^{8} (2)(4)d\lambda = 8\lambda\Big|_{t-6}^{8} = 112 - 8t$$

Hence,

z(t) = **8t, 0>t>2**

16–8t, 2>t>6

–16, 6>t>8

8t–80, 8>t>12

112–8t, 12>t>14

0, otherwise.